

ENTERPRISE INFRASTRUCTURE PLAN

NexaForge Software Solutions

Week 2 Portfolio Project

ITEP 414 - System Administration and Maintenance

Bachelor of Science in Information Technology (BSIT)

Prepared by: Renz Rouei Villegas

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Office: Santa Cruz, Laguna, Philippines (fictional office)

Executive Summary

NexaForge Software Solutions is a fictional 20-employee software development startup composed of Information Technology (5), Human Resources (4), Finance (5), and Sales (6). This plan designs its initial IT environment from scratch, covering company requirements, hardware, software, networking, staff roles, security, backup, and growth. The design favors practical business-class equipment, a managed star-topology LAN, Linux server capability, documented administration practices, and security controls that can scale with the company.

1. Company Profile

Company Name: NexaForge Software Solutions

Nature of Business: Custom software development, web/mobile application development, cloud integration, and technical consulting for small and medium-sized businesses.

Vision: To become a trusted Philippine technology partner known for reliable, secure, and practical software solutions that help organizations modernize their operations.

Mission: To design and deliver maintainable digital solutions while building a secure, collaborative, and continuously improving workplace for clients and employees.

Office Location: Santa Cruz, Laguna, Philippines (fictional office)

Organizational Structure: Managing Director -> IT / Development, Human Resources, Finance, and Sales. The Junior System Administrator operates under IT and supports all departments.

Department	Employees	Primary IT Needs
Information Technology	5	Development workstations, server administration, Git, virtualization, privileged access.
Human Resources	4	Office productivity, secure personnel records, printing, controlled file access.
Finance	5	Reliable desktops, secure financial data, MFA, backups, printing.
Sales	6	Mobile laptops, Wi-Fi, browser/office tools, remote access, client presentations.
TOTAL	20	Shared network, internet, server, printing, backup, security, support.

2. Enterprise Hardware Inventory

The inventory uses realistic quantities for a 20-person startup while preserving spare capacity for shared devices and future growth.

Asset ID	Hardware	Qty	Department	Purpose / Justification
NF-WS-001	Desktop Computers	14	IT (4), HR (4), Finance (4), Sales (2)	Primary fixed workstations; strong value and easier maintenance.
NF-LT-001	Business Laptops	8	IT (1), Finance (1), Sales (6)	Mobility for client meetings, remote work, and administrative tasks.
NF-SRV-001	Business Server	1	IT / Server Room	Hosts internal file sharing, development services, authentication-ready workloads, and backups.
NF-RTR-001	Business Router	1	Network Core	WAN routing, DHCP, VPN capability, and traffic management.
NF-SW-001	48-Port Managed Gigabit Switch	1	Network Core	Enough ports for 20 staff plus server, printer, APs, NAS, and growth.
NF-PRN-001	Network Multifunction Printer	2	Shared	General printing plus a unit near HR/Finance for controlled document workflows.
NF-UPS-001	1500VA UPS	2	Server/Network Core	Keeps critical devices online during short outages and supports safe shutdown.
NF-AP-001	Wi-Fi 6 Access Point	2	Office-wide	Provides overlapping staff wireless coverage and capacity.
NF-NAS-001	4-Bay NAS	1	IT / Server Room	Centralized local backup and shared storage with RAID support.
NF-BKP-001	8 TB External Backup Drive	2	IT	Rotated/offline backup copies; one can be stored off-site.
NF-MON-001	24-inch Monitors	22	All Departments	One per user plus two extra units for IT troubleshooting/spares.

Selection Rationale

Fourteen desktops cover staff who primarily work at fixed desks, while eight laptops support the six-person Sales team plus mobile needs in IT and Finance. A 48-port managed switch is intentionally larger than the immediate endpoint count so the company can add phones, cameras, access points, test devices, and new employees without replacing the core switch. The server, NAS, UPS units, and rotated external drives provide a basic foundation for internal services and recovery.

3. Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current supported release	Commercial / OEM	Business endpoints with organizational management and security features.
Ubuntu Server 26.04 LTS	26.04 LTS	Open source; optional Ubuntu Pro	Internal Linux services and future web/database workloads.
Microsoft 365 Apps	Current subscription	Commercial subscription	Documents, spreadsheets, presentations, email, and collaboration.
Visual Studio Code	Current stable	Free	Code and configuration editor.
Git	Current stable	GPLv2	Distributed version control for code, scripts, and documentation.
GitHub Desktop	Current stable	Free	Graphical Git client for repository management.
Oracle VirtualBox	Current stable	GPL / PUUEL components	Virtualization for testing operating systems and lab environments.
Google Chrome	Current stable	Freeware	Web browser for business and development tools.
Microsoft Defender	Built into Windows 11	Included with Windows	Endpoint anti-malware and baseline Windows protection.
AnyDesk	Current stable	Commercial for business use	Authorized remote troubleshooting and support.
7-Zip	Current stable	Open source	File compression and archive management.

Ubuntu Server is selected for the internal server because an LTS release provides a stable Linux base suitable for file, web, database, and development services. Windows 11 Pro is used on business endpoints. Development and administration tools such as VS Code, Git, GitHub Desktop, and VirtualBox support the company's software-development workflow and testing needs.

4. Enterprise Network Inventory

Equipment	Quantity	Purpose
ISP Modem/ONT	1	Terminates business internet service and hands off Ethernet to the router.
Business Router	1	WAN/LAN routing, DHCP, VPN capability, and traffic management.
Next-Generation Firewall	1	Traffic filtering, VPN, segmentation, threat controls, and security logging.
48-Port Managed Gigabit Switch	1	Core wired connectivity with VLAN, QoS, and monitoring support.
Wi-Fi 6 Access Point	2	Business wireless coverage plus a logically separated guest network.
24-Port Patch Panel	2	Structured cabling termination and easier rack maintenance.
CAT6 Cable	2 boxes / as required	Gigabit-capable horizontal cabling.
RJ45 Connectors / Keystones	50+	Cabling termination and spares.

Network Design Notes

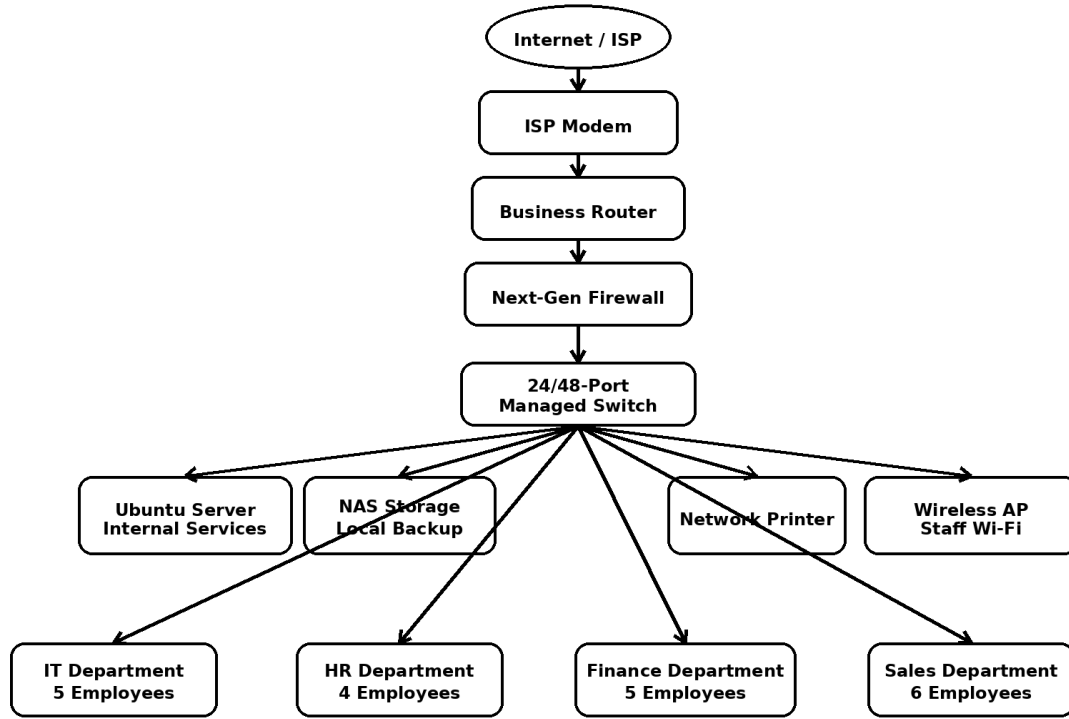
The office uses a logical star topology centered on a managed switch. Internet traffic passes through the ISP modem/ONT, business router, and dedicated firewall before reaching the LAN. The firewall and managed switch allow future VLAN separation for staff, guest Wi-Fi, servers, and network management. Wired connections are preferred for fixed desktops, the server, NAS, and printer; Wi-Fi is primarily for laptops and mobile devices.

5. Enterprise Network Diagram

The following diagram includes the required Internet, ISP modem, router, firewall, switch, wireless access point, server, printer, and all four departments.

NexaForge Software Solutions - Enterprise Network Topology

Prepared by: Renz Rouei Villegas



Logical star topology: Internet -> modem -> router -> firewall -> managed switch -> servers, shared devices, access point, and departments.

Figure 1. Proposed enterprise network topology.

6. System Administration Roles

Helpdesk Technician

Responsibilities: Receives user incidents and service requests, troubleshoots client hardware/software, performs basic account and endpoint support, documents solutions, and escalates complex issues.

Skills: Customer communication, Windows troubleshooting, basic networking, ticket documentation, remote support, hardware diagnosis, and security awareness.

Common Tools: Ticketing system, Remote Desktop/AnyDesk, Microsoft Defender, Event Viewer, command-line utilities, knowledge base, asset inventory.

Certifications: CompTIA A+ is a common entry-level choice; Microsoft fundamentals credentials can complement Windows support skills.

Network Administrator

Responsibilities: Plans, configures, monitors, and maintains LAN/WAN services, switches, routers, wireless access, VLANs, VPNs, addressing, and network security controls.

Skills: TCP/IP, subnetting, switching/routing, VLANs, wireless, firewall concepts, monitoring, troubleshooting, documentation, and automation basics.

Common Tools: Managed switch/router interfaces, Wireshark, ping/traceroute, SSH, SNMP/network monitoring, configuration backups.

Certifications: Cisco CCNA aligns with network fundamentals, network access, IP connectivity/services, security fundamentals, and automation.

Linux System Administrator

Responsibilities: Installs and maintains Linux servers, manages users and permissions, packages, services, storage, backups, logs, SSH, updates, and baseline security.

Skills: Linux CLI, Bash, systemd, filesystems, permissions, networking, package management, logs, backup/recovery, and scripting.

Common Tools: SSH, Bash, systemctl, journalctl, apt/dnf, cron/systemd timers, rsync, Git, monitoring tools.

Certifications: Red Hat Certified System Administrator (RHCSA) validates core Linux system administration skills.

Cloud Administrator

Responsibilities: Implements and monitors cloud identities, virtual networks, storage, compute, security, governance, backup, and cost-aware resource operations.

Skills: Cloud networking, IAM, virtual machines, storage, monitoring, security, governance, scripting, and hybrid infrastructure concepts.

Common Tools: Azure portal/CLI, Microsoft Entra ID, Azure Monitor, Azure Backup, PowerShell, infrastructure documentation.

Certifications: Microsoft Certified: Azure Administrator Associate (AZ-104) aligns with identities/governance, storage, compute, virtual networking, and monitoring.

How the Four Roles Work Together

Inside one organization, these roles form an escalation and operations chain. The Helpdesk Technician handles frontline user issues and records recurring problems. Network Administrators maintain connectivity. Linux System Administrators maintain server operating systems, services, accounts, storage, updates, and recovery. Cloud Administrators extend the environment into cloud identity, compute, storage, networking, monitoring, and governance. They share documentation and monitoring information, escalate incidents to the appropriate specialist, coordinate changes, and apply consistent security policies so endpoint, network, server, and cloud components operate as one environment.

7. Infrastructure Recommendations

Internet Provider

Use a reputable Philippine business-fiber provider that offers at least 300-500 Mbps service, business support, and an SLA where available. The final provider should be selected after checking serviceability at the fictional office. Add a secondary connection when the business becomes dependent on continuous cloud access.

Server Specifications

Use one business-class server with an 8-core server-grade CPU, 32 GB ECC RAM expandable to 64 GB or more, two 2 TB enterprise SSDs in RAID 1, dual network interfaces, TPM, remote-management capability, and backup to the NAS. This is intentionally moderate for a 20-person startup and leaves room for internal services and light virtualization.

Backup Strategy

Apply the 3-2-1 approach: production data plus two backup copies, on at least two media types, with one copy off-site. Use nightly automated NAS backups, weekly encrypted external-drive rotation, and a vetted cloud/off-site copy for critical data. Test restoration regularly and keep at least one backup offline or isolated.

Security Recommendations

Place a dedicated firewall between the router and internal switch; separate staff, guest Wi-Fi, and server/management traffic with VLANs; enable MFA; patch systems; encrypt endpoints; use least privilege; log administrative activity; and maintain an asset inventory.

Antivirus

Use Microsoft Defender on Windows endpoints as the baseline and centrally monitor protection status where licensing permits. Combine anti-malware with patching, least privilege, browser protection, and user awareness.

Password Policy

Require unique passwords/passphrases, block known compromised/default credentials, use a password manager, enable MFA wherever available, and require stronger controls for administrators and Finance. Do not rely on password changes alone as a security control.

Expansion Plan

Reserve switch capacity and IP space for at least 30-40 users, document VLANs and addressing, use modular server/NAS storage, and review infrastructure every six months. As the company grows, add a second ISP, additional APs, centralized identity/device management, and cloud disaster-recovery options.

8. Personal Reflection

Completing this infrastructure planning project helped me understand that system administration begins long before a server is installed or a cable is connected. The most important lesson I learned was that hardware, software, networking, security, documentation, and business requirements have to be planned as one system. A decision in one area affects the others. For example, choosing a managed switch is not only a hardware decision because it also supports network segmentation, future expansion, troubleshooting, and security.

The most challenging part was balancing the needs of four departments while keeping the design realistic for a startup with only twenty employees. It would have been easy to overdesign the environment with unnecessary enterprise equipment, or underdesign it and leave no room for growth. I had to think about which employees need fixed desktops, which roles benefit from laptops, how many switch ports are required, how backups should be separated, and how the server and network devices should be protected.

Planning before deployment is important because mistakes are much cheaper to correct on paper than after equipment has been purchased and configured. A clear infrastructure plan also gives administrators a reference for troubleshooting, upgrades, budgeting, and onboarding future IT staff. Documentation reduces guesswork and makes decisions easier to explain to management.

This project will help me become a better System Administrator because it trained me to think beyond individual devices. Instead of only asking whether a computer or server works, I now have to consider reliability, security, maintainability, scalability, and the needs of the people using the system. It also reinforced the importance of documenting technical decisions so another administrator can understand and continue the work.

Word count: 274 words.

References

The course module is the basis for the required project structure. External references support the technical role, certification, Ubuntu Server, backup, and cybersecurity recommendations.

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